A CONTINUOUS FUNCTOR WITH FILTERED AUXILIARY CATEGORIES DEFINES A MORPHISM OF SITES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Exactness of left adjoints. Categories, Lemma categories-lemma-exact-adjoint. Gives right exactness of the induced functor on sheaves.
- Filtered colimits commute with finite limits. Categories, Lemma categories-lemma-directed-commutes together with the description of limits in presheaves. Shows that the left Kan extension commutes with finite limits.
- Sheafification commutes with finite limits. Same chapter, Lemma lemma-sheafification-exact. Transfers the commutation from presheaves to sheaves.

Source reference. Original section 00VC: sources/stacks-sites/pinned-original-source.tex, source lines 3023–3023. Navigation pointer invoked at proof line 3023; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0039: sources/stacks-sites/pinned-original-source.tex, source lines 2975–2975. Navigation pointer invoked at proof line 2975; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 002W: sources/stacks-sites/pinned-original-source.tex, source lines 3035–3035. Navigation pointer invoked at proof line 3035; the counted

Date: stacks-project:sites.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original section 00VB: sources/stacks-sites/pinned-original-source.tex, source lines 3037–3037. Navigation pointer invoked at proof line 3037; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00WJ: sources/stacks-sites/pinned-original-source.tex, source lines 2972–2972. Navigation pointer invoked at proof line 2972; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ and $\mathcal{D}$ be sites. Let $u : \mathcal{C} \to \mathcal{D}$ be continuous. Assume all the categories $(\mathcal{I}_V^u)^{opp}$ of Section 00VC are filtered. Then u defines a morphism of sites $\mathcal{D} \to \mathcal{C}$, in other words u_s is exact.*

Proof. Since u_s is the left adjoint of u^s we see that u_s is right exact, see Categories, Lemma 0039. Hence it suffices to show that u_s is left exact. In other words we have to show that u_s commutes with finite limits. Because the categories $\mathcal{I}_Y^{opp}$ are filtered we see that u_p commutes with finite limits, see Categories, Lemma 002W (this also uses the description of limits in *PSh*, see Section 00VB). And since sheafification commutes with finite limits as well (Lemma 00WJ) we conclude because $u_s = \# \circ u_p$. □